Inertial Particles and Turbulence

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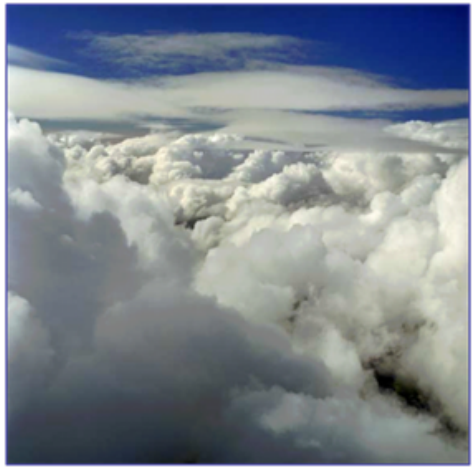


November 25th 2026 - Chatou

Inertial particles and turbulence

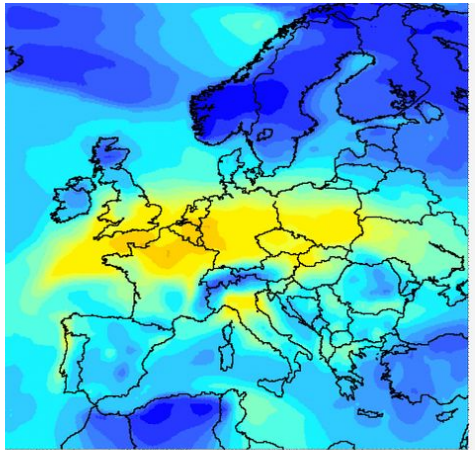










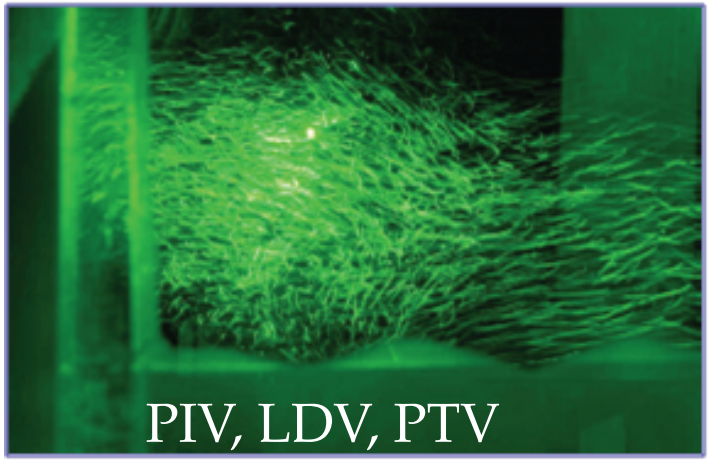




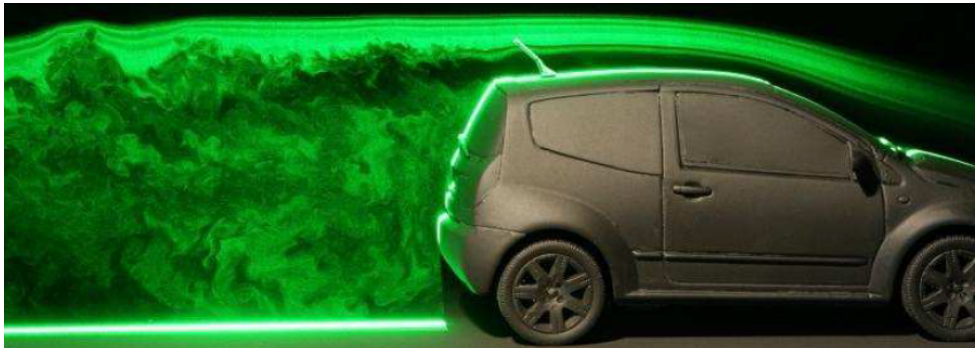


Tracers *vs* Inertial Particles

Tracers behave as fluid particles



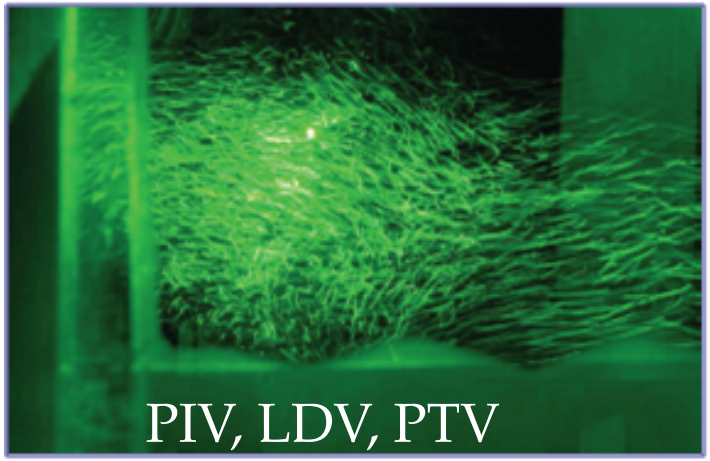
Inertial particles do not follow the flow



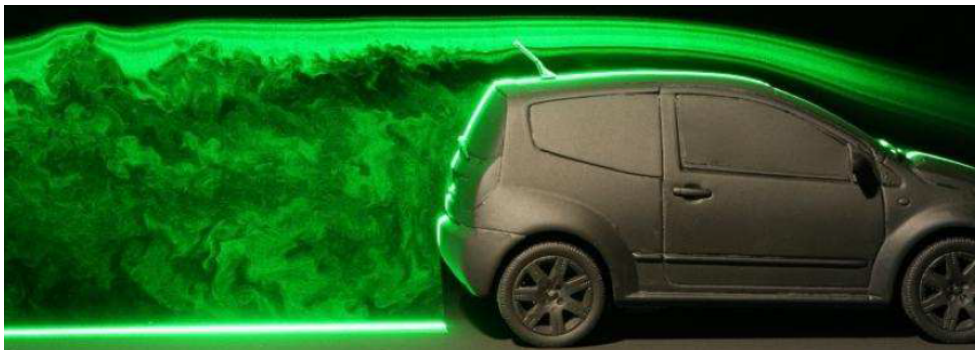
Grandemange et al, ENSTA 2016

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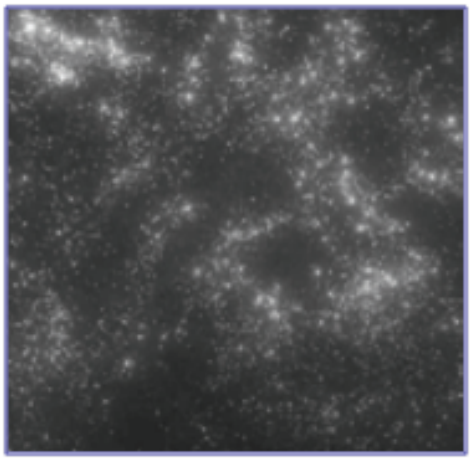


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Grandemange et al, ENSTA 2016

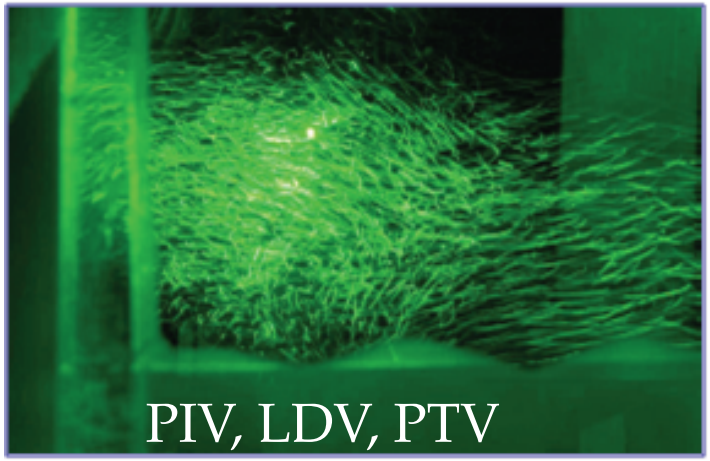
Cluster formation



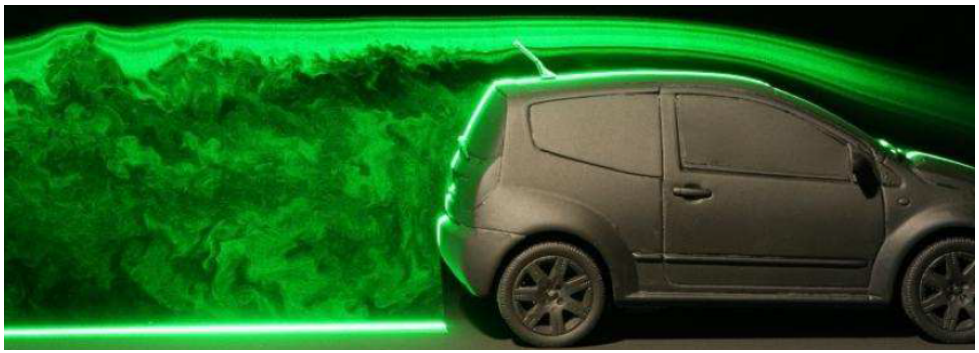
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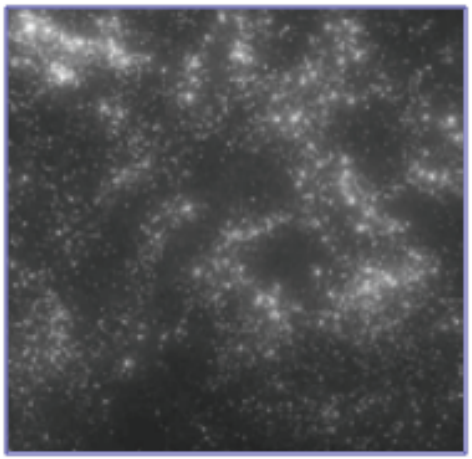


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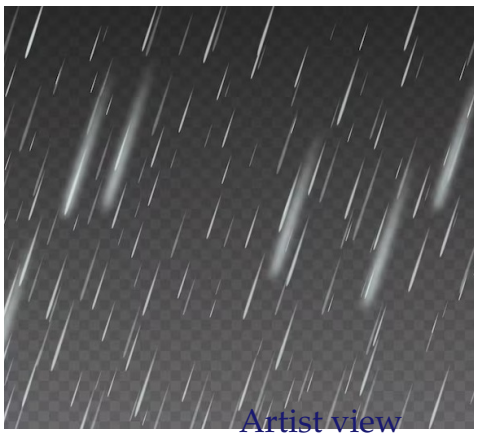
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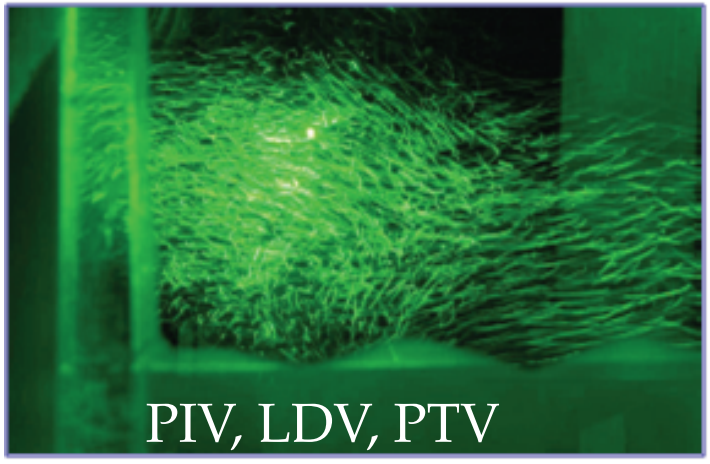
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Settling alteration

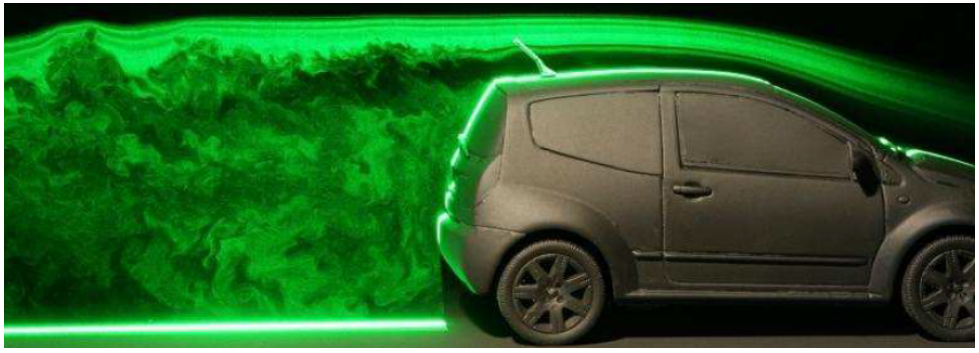


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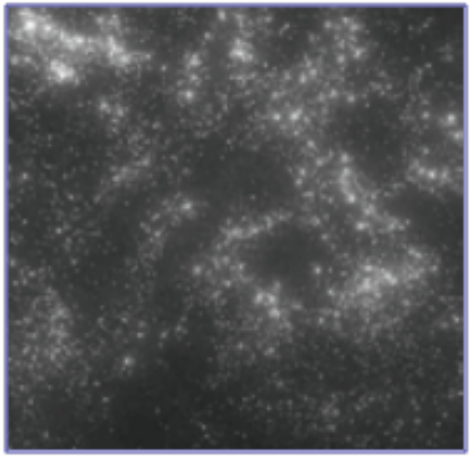


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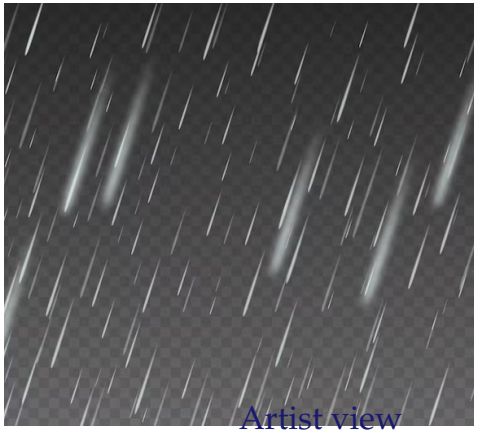
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Settling alteration



Artist view

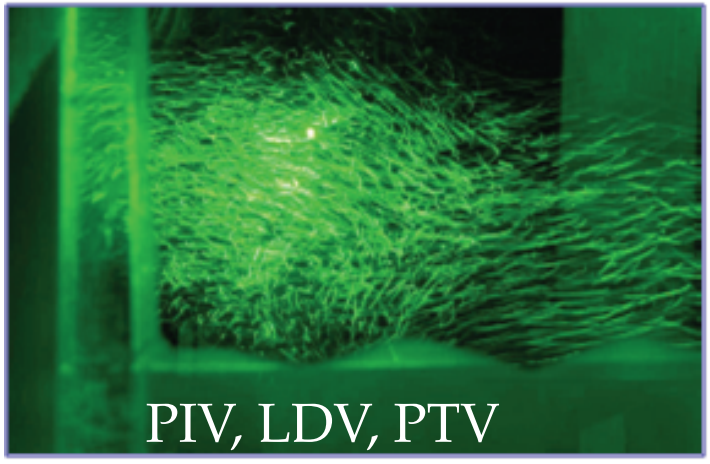
Collision alteration



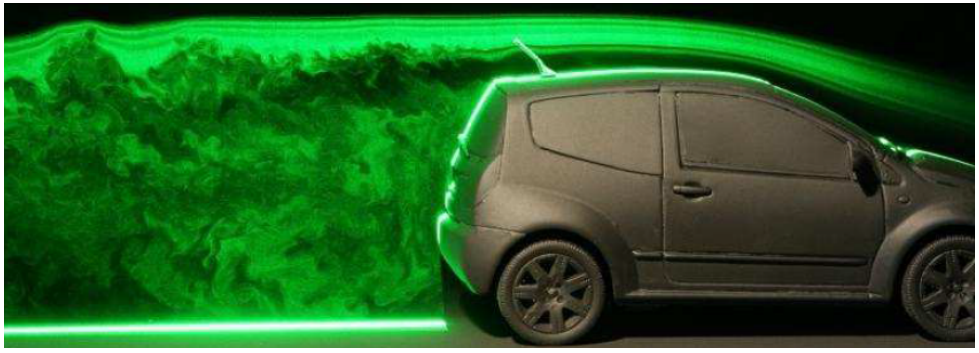
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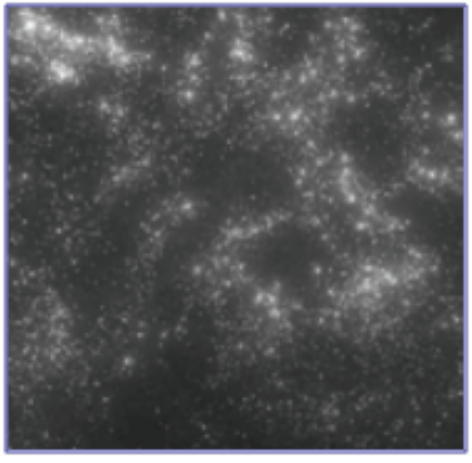


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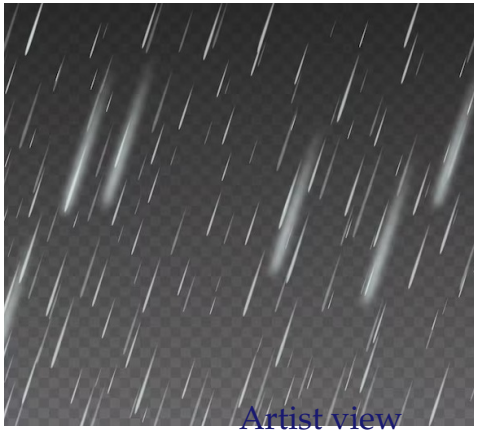
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Artist view

Collision alteration



Artist view

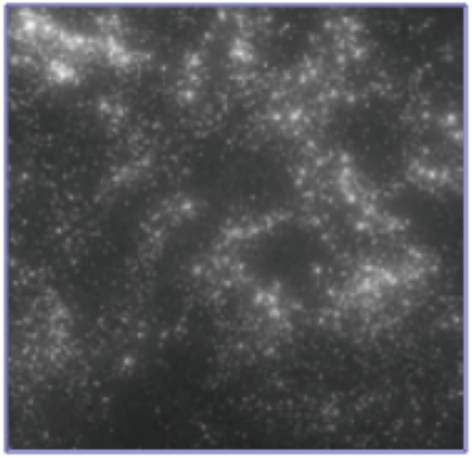
Flow modulation, Dispersion...





Increase meeting probabilities

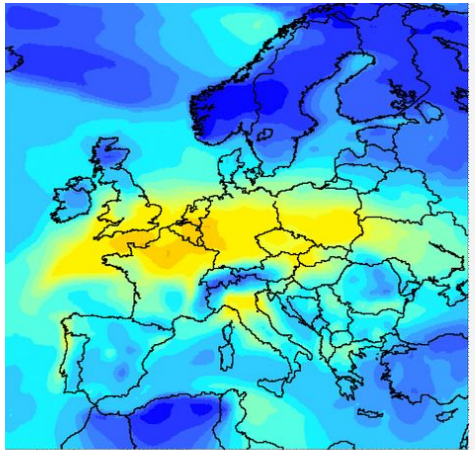
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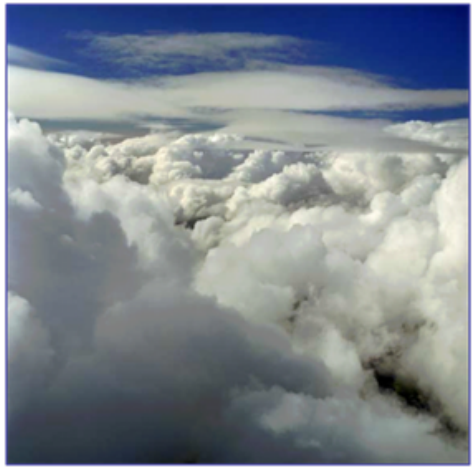
Stoichiometric proportion



Sanitary thresholds







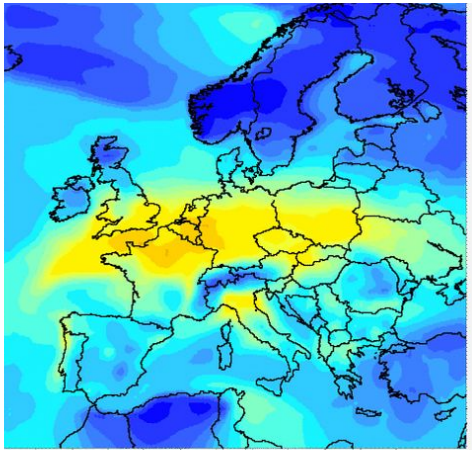
Settling alteration

-

Dispersion

Rain formation alteration

Aerosol and climate forcing
Aviation safety



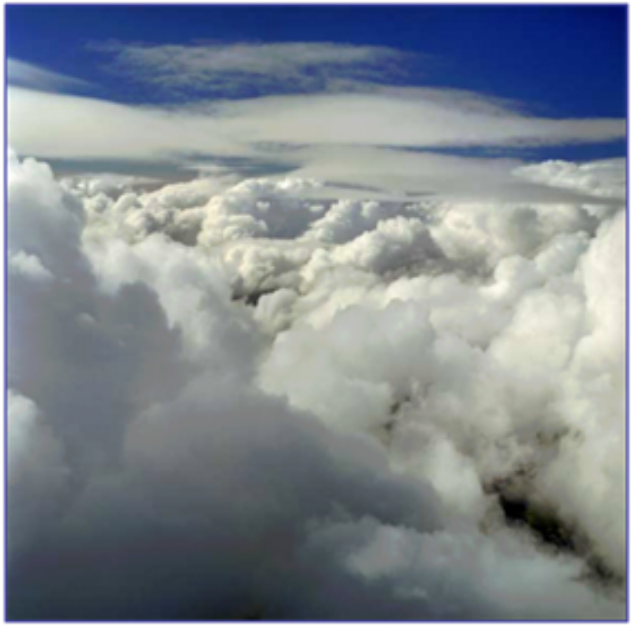




Marine snow : carbon pump

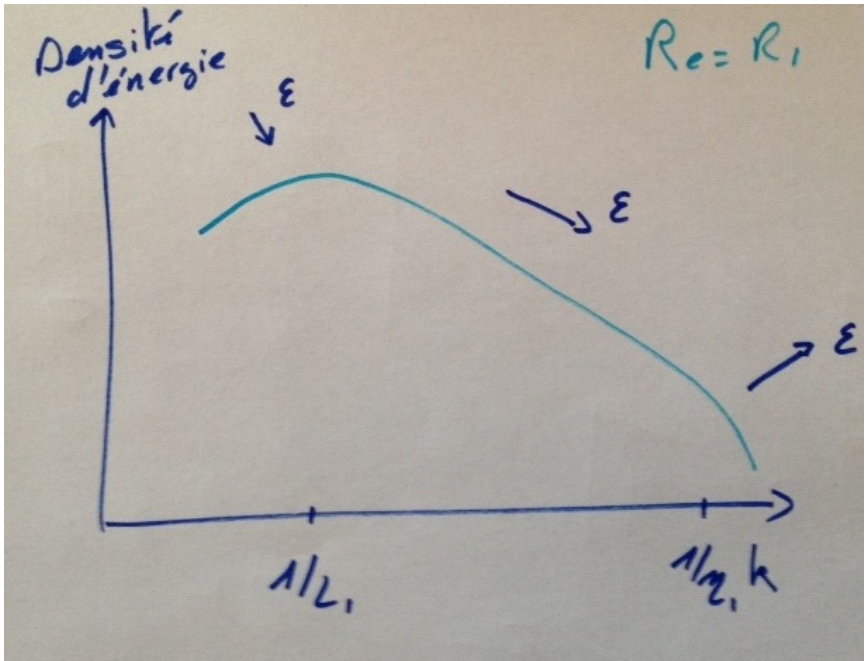
Estuaries dynamics

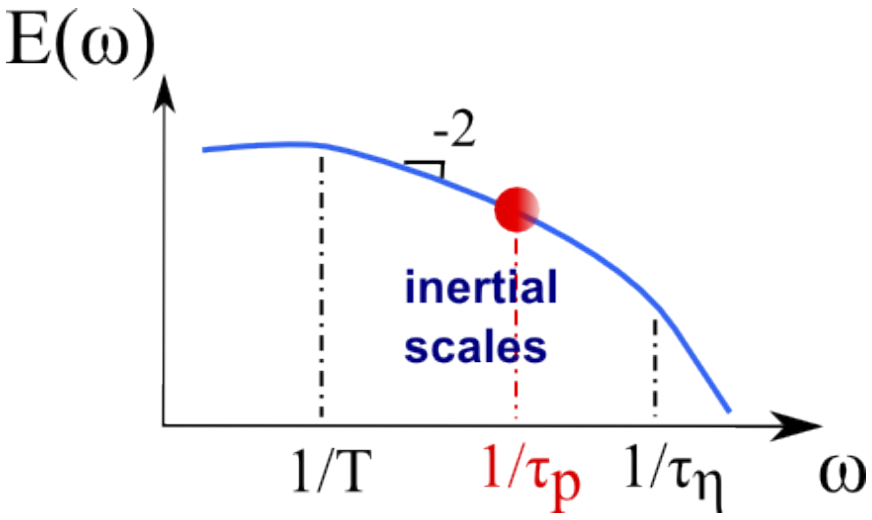


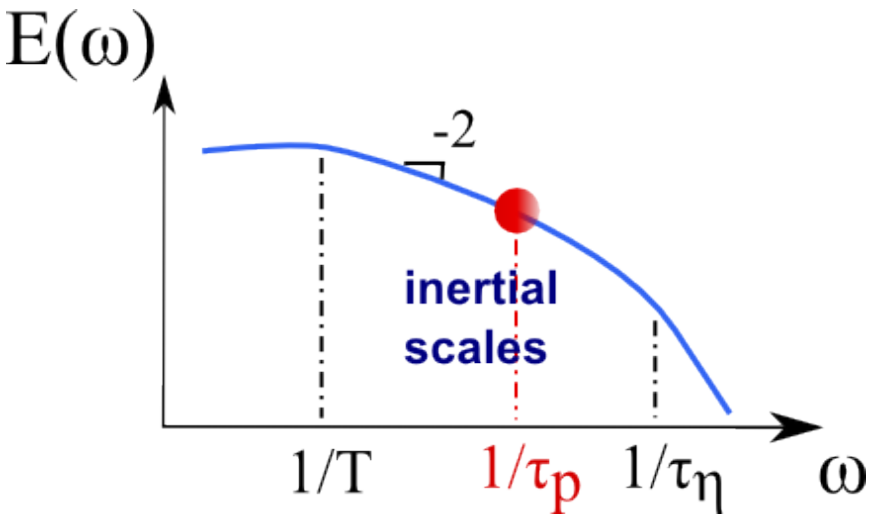







Densité d'énergie
ε
ε
ε
$Re = R_1$
$1/L_1$
$1/\eta_1 k$


E(ω)
-2
inertial
scales
1/T
1/τp
1/τη
ω

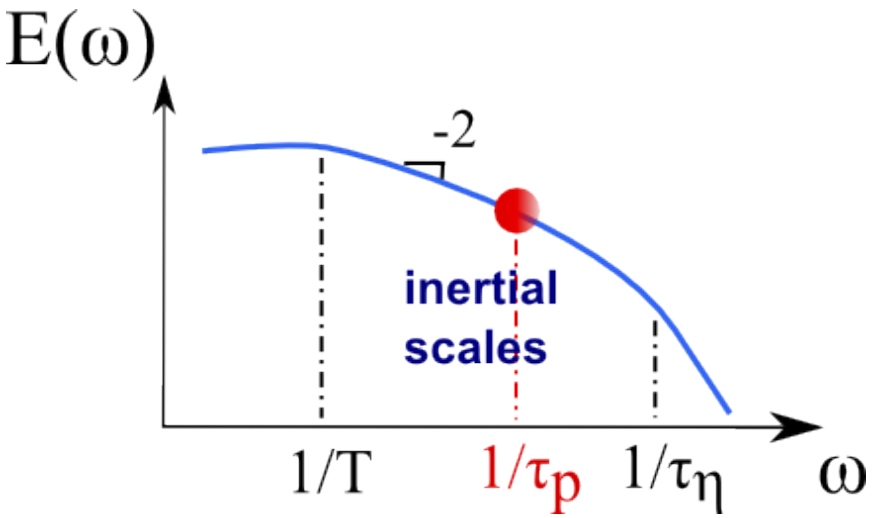


Model point particle equation :

$$\rho_p \frac{\partial v_i}{\partial t} = \overbrace{(\rho_p - \rho_f) g_i}^{\text{gravity}} + \overbrace{\frac{\partial \tau_{ij}}{\partial x_j} - \frac{\partial p}{\partial x_i}}^{\text{undisturbed flow}} + \overbrace{\frac{9}{2}\frac{\mu}{r_p^2}\left((u_i - v_i) + \frac{r_p^2}{6}\nabla^2 u_i\right)}^{\text{steady state drag}}$$

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Maxey & Riley, *POF*, 26, **1983**
Gattignol, *JMTA*, 2, **1983**



Response time :

Stokes drag : $\tau_p = \rho_p d_p^2 / 18\mu$

Non linear drag : $\tau_p^{eff}(|\mathbf{w}|)$

Added mass : τ_p^{am}

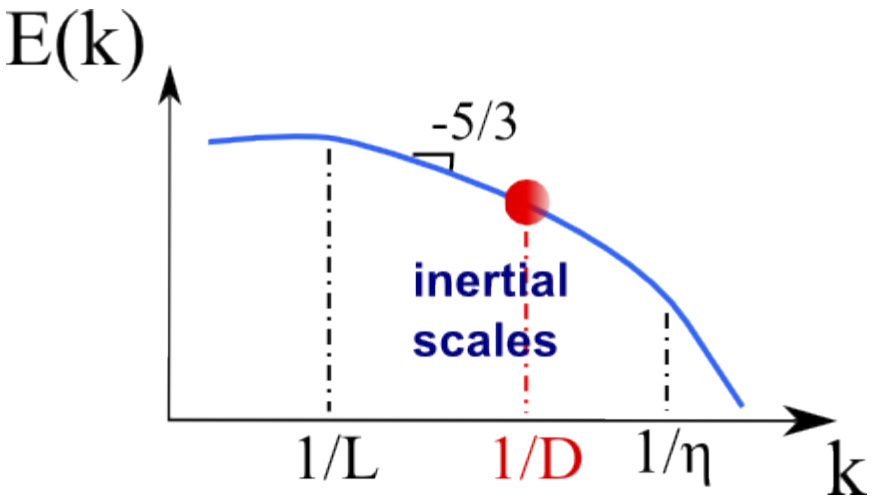
Basset history : ...

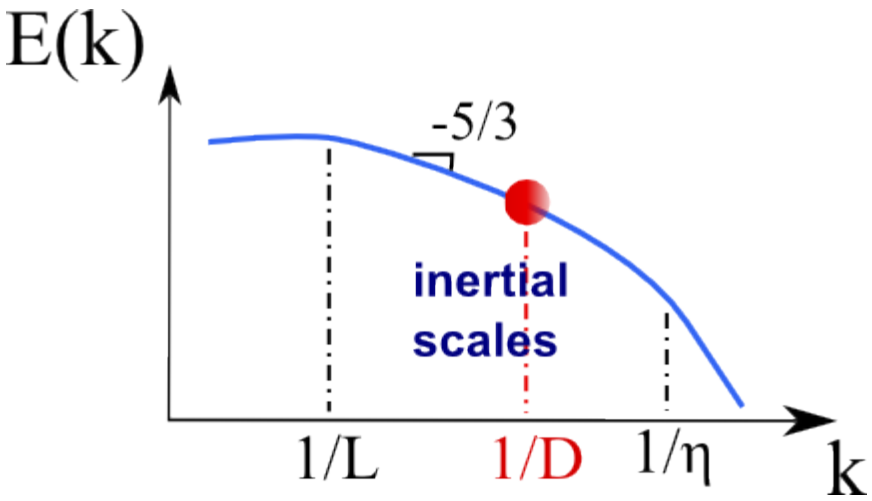
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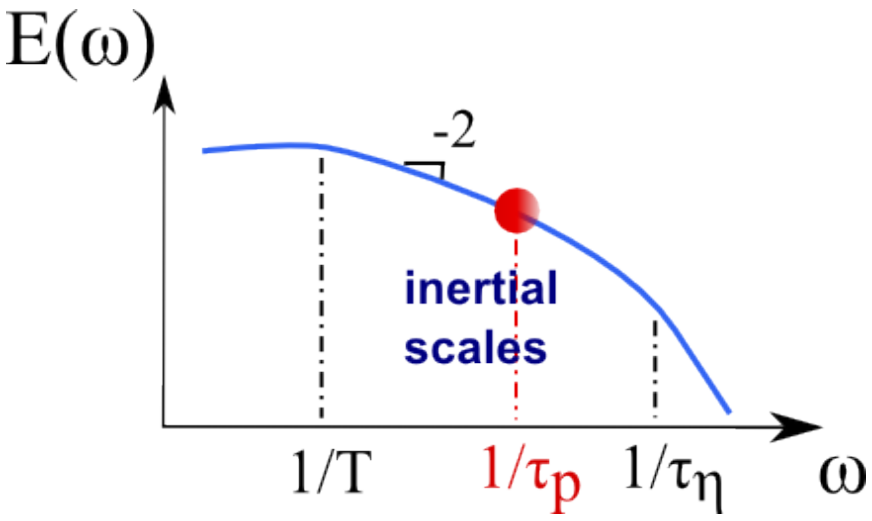
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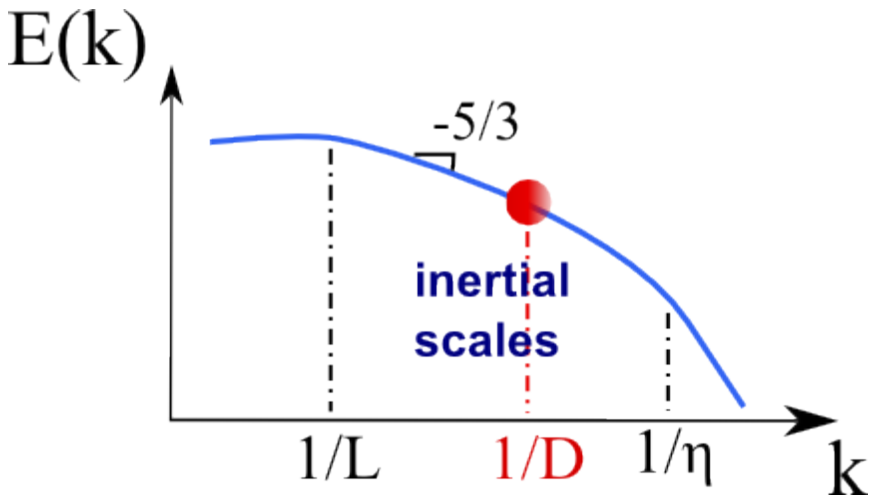

E(k)
-5/3
inertial
scales
1/L
1/D
1/η
k



Finite size effects :

- Faxén corrections ($d_p \leq \eta$)
- Filtering of small scales, integration over particle size
- Anisotropy
  - Rotation / tumbling inertia
  - Porosity and roughness
- ...





Source	**Timescale**	**Relevant when**
Stokes drag	$\tau_p = \rho_p d_p^2/18\mu$	Always, baseline
Nonlinear drag	$\tau_p^{eff}(\lvert\mathbf{w}\rvert)$	$Re_p \gtrsim 1$
Added mass	τ_p^{am}	$\rho_p/\rho_f \sim \mathcal{O}(1)$
Basset history	complex kernel	Unsteady $\mathbf{w}$, ρ_p/ρ_f moderate
Orientational inertia	τ_{rot}	Non-spherical particles
Finite size filtering	d_p/η	$d_p \gtrsim \eta$
Porosity/permeability	Internal flow timescale	Aggregates, flocs

Control parameters

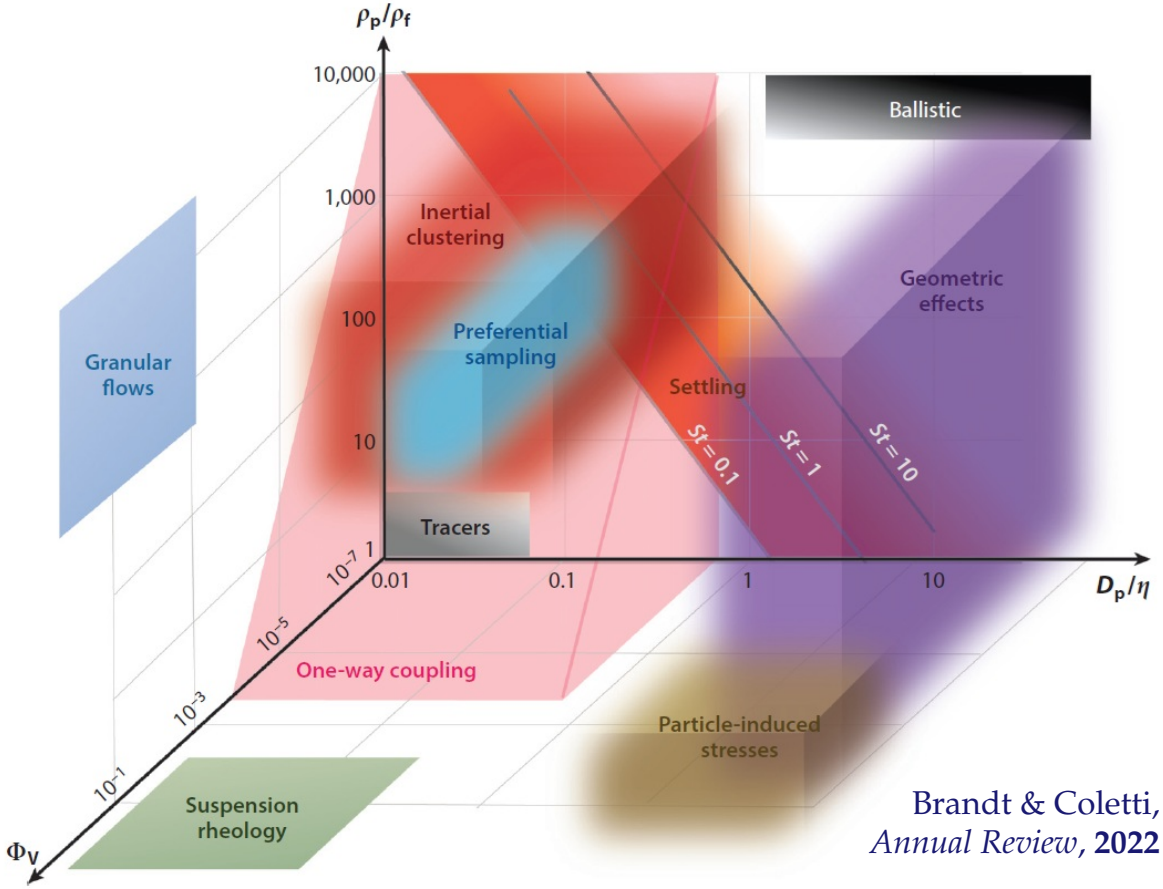
Reynolds number	Stokes number	Rouse number	Density ratio	Loading
$\mathbf{R}_\lambda = \frac{u'\lambda}{\nu}$;	$\mathbf{St} = \frac{2}{9}\left(\frac{r_p}{\eta}\right)^2\left(\frac{\rho_p}{\rho_f} - 1\right)$;	$\mathbf{R} = \frac{v_t}{u'}$;	$\Gamma = \frac{\rho_p}{\rho_f}$;	Φ_v.
Turbulence	**Inertia**	**Gravity**	**Density**	**Retroaction**
$10 - 10^4$	$10^{-5} - 10^2$	$0 - 10$	$10^{-4} - 10^3$	$10^{-6} - 10^{-1}$

- Particle Reynolds number
- Galileo number
- Aspect ratio
- Deformability
- ...

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Brandt & Coletti, *Annual Review*, **2022**

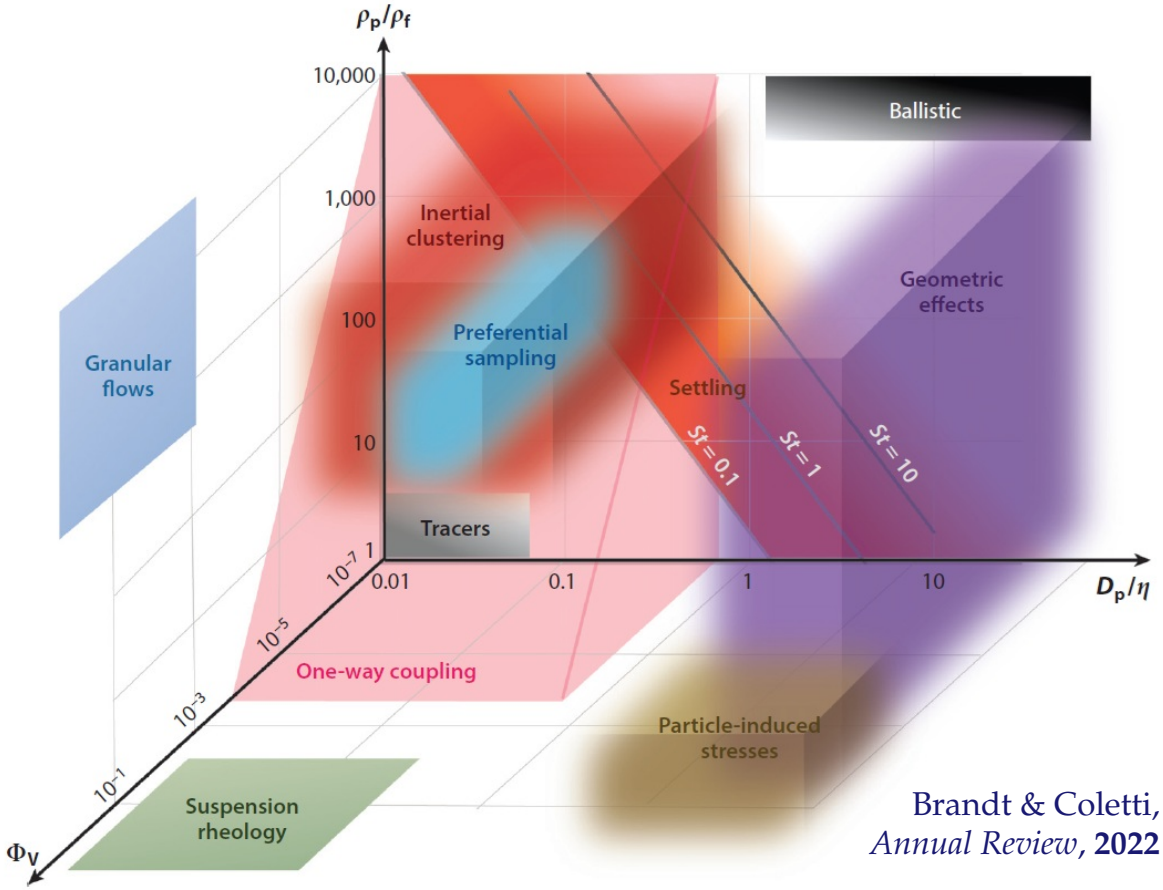
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Intertwined parameters

$\Longrightarrow$ complementary simulations and experiments



Brandt & Coletti, *Annual Review*, **2022**

Point particle approaches

- Central assumption : ***undisturbed flow velocity***
- Quasi-steady forces : drag (Stokes / nonlinear), lift
- Unsteady forces : added mass, pressure gradient, Basset
- Finite-size correction : Faxén : $\mathbf{u}_{eff} = \mathbf{u}(\mathbf{x}_p) + \frac{d_p^2}{24}\nabla^2\mathbf{u}(\mathbf{x}_p)$
- Validity : $d_p \ll \eta$, $Re_p \ll 1$

Particle equation (Maxey-Riley-Gattignol) :

$$\rho_p \frac{\partial v_i}{\partial t} = \overbrace{(\rho_p - \rho_f)g_i}^{\text{gravity}} + \overbrace{\frac{\partial \tau_{ij}}{\partial x_j} - \frac{\partial p}{\partial x_i}}^{\text{undisturbed flow}} + \overbrace{\frac{9}{2}\frac{\mu f_D}{r_p^2}\left((u_i - v_i) + \frac{r_p^2}{6}\nabla^2 u_i\right)}^{\text{steady state drag}}$$

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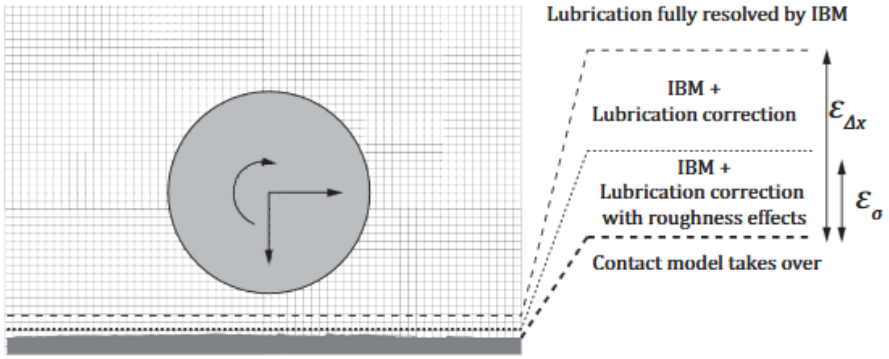
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Coupling :

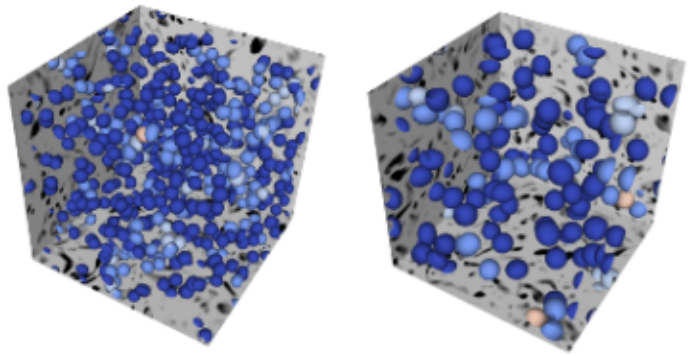
- 1-way coupling : fluid a priori + particle equation
- 2-way coupling : Navier-Stokes with particle back reaction
- ”4-way” coupling : inter particle momentum exchange
  - collisions
  - near field approaches
  - Lagrangian framework

Resolved particle approaches

- Finite size particles
- Key advantages :
  - exact surface stress
  - exact drag
  - no undisturbed velocity needed
- Key limitations :
  - computational cost
  - fewer particles
  - lower R_λ



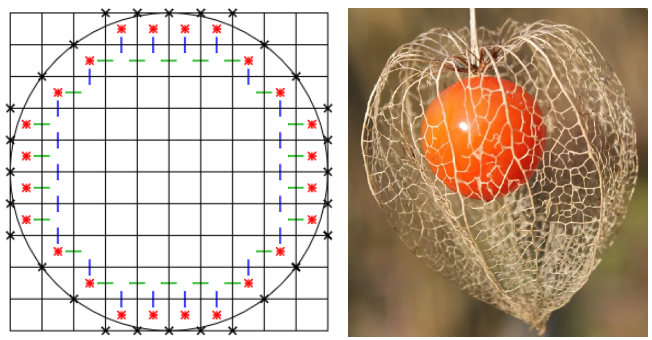
IBM : Costa et al, PRE, 2015



Basilisk : Boniou et al, JFM, 2025

A wide range of methods :

- Immersed Boundary Method
- Force Coupling Method
- Lattice Boltzmann
- Basilisk
- Physalis
- …

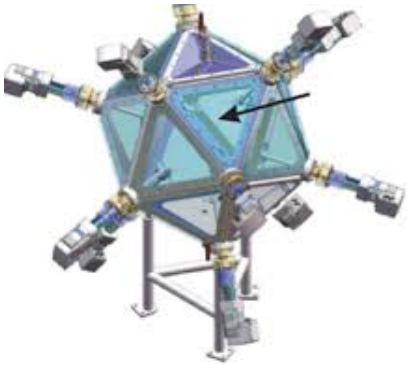


Physalis : Naso & Prosperetti, NJP, 2010

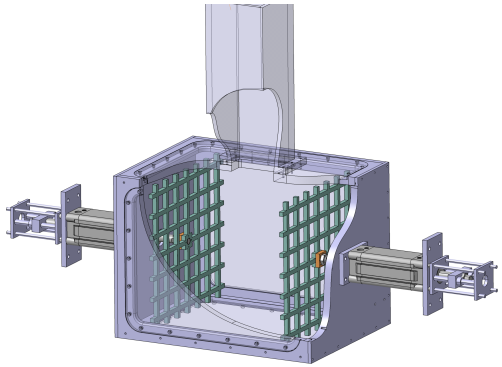
See review by Maxey, ARFM, 2017

Turbulent generators

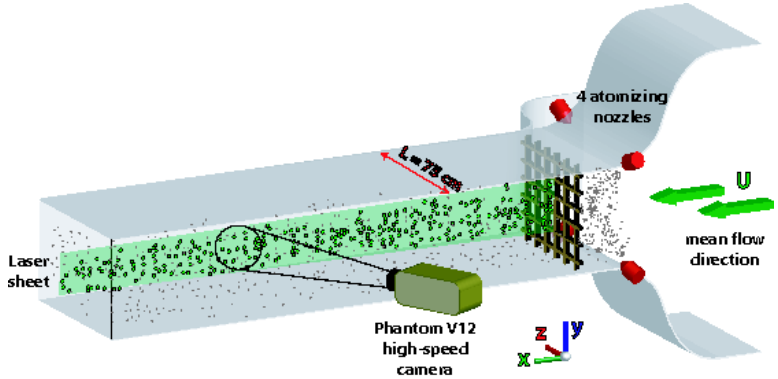
- Wind tunnels :
  - passive or active grids
  - *good homogeneity and isotropy*
  - *strong mean flow*
- Turbulent boxes :
  - von Kármán
  - polyhedra with fans / motors
  - oscillating grids
  - synthetic jets
  - *vanishing mean low*
  - *recirculation problem*
- In situ measurements



LEM : Zimmerman et al, RSI, 2010



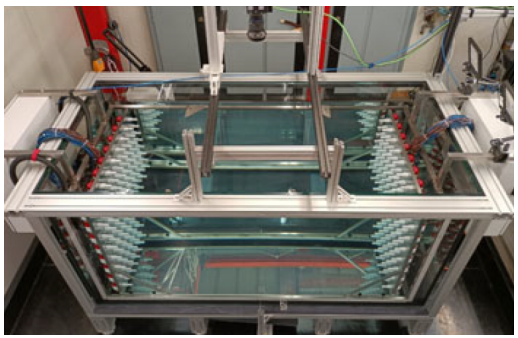
Oscillating grids : ENSTA



Wind tunnel : Monchaux et al, POF, 2009

Trade off :

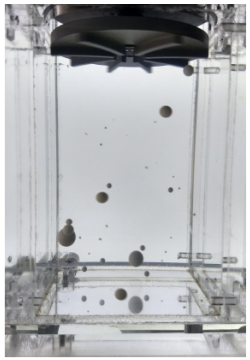
- Homogenity and sotropy
- Turbulence level R_λ

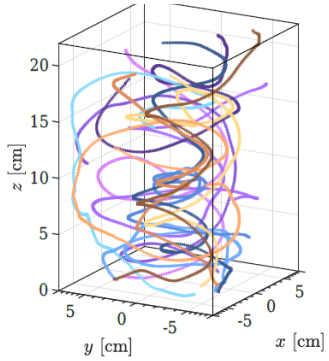


Jet arrays : Coletti group, Zurich

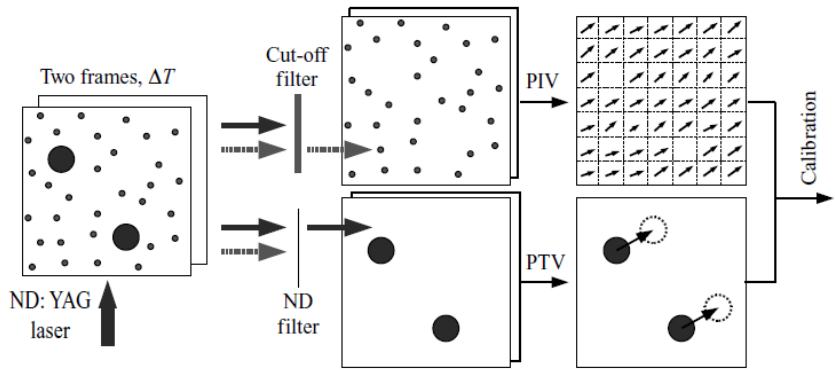
Turbulence measurements

- 2D : PIV, PTV
- 3D : Tomo-PIV, Holographic PIV, 3D-PTV
- Simultaneous fluid+particle :
  - fluorescence separation
  - refractive index matching
  - scale separation
- Opaque flows :
  - X-ray PIV
  - acoustic Doppler





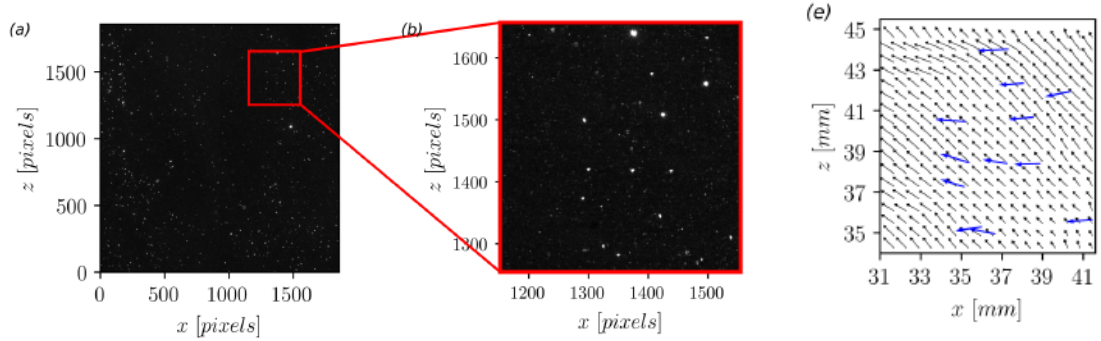
ENS Lyon group



Poelma et al, JFM, 2007

Seeding :

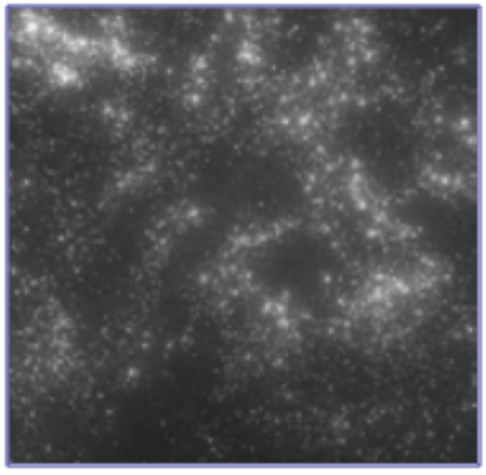
- Particle characterization
- Polydispersity
- Seeding homogeneity
- Disentangling parameters



Coletti group, Zurich

Main features

Cluster formation



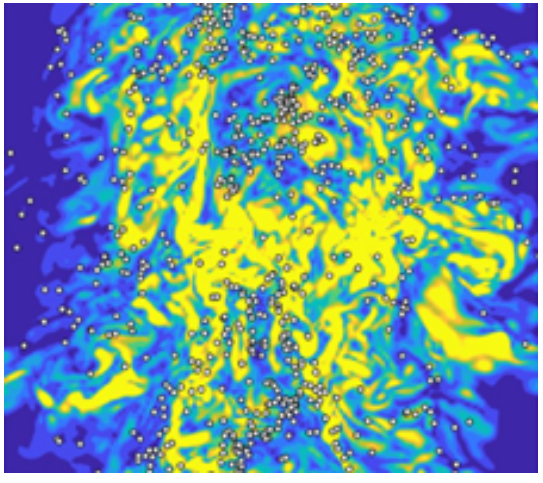
Settling alteration



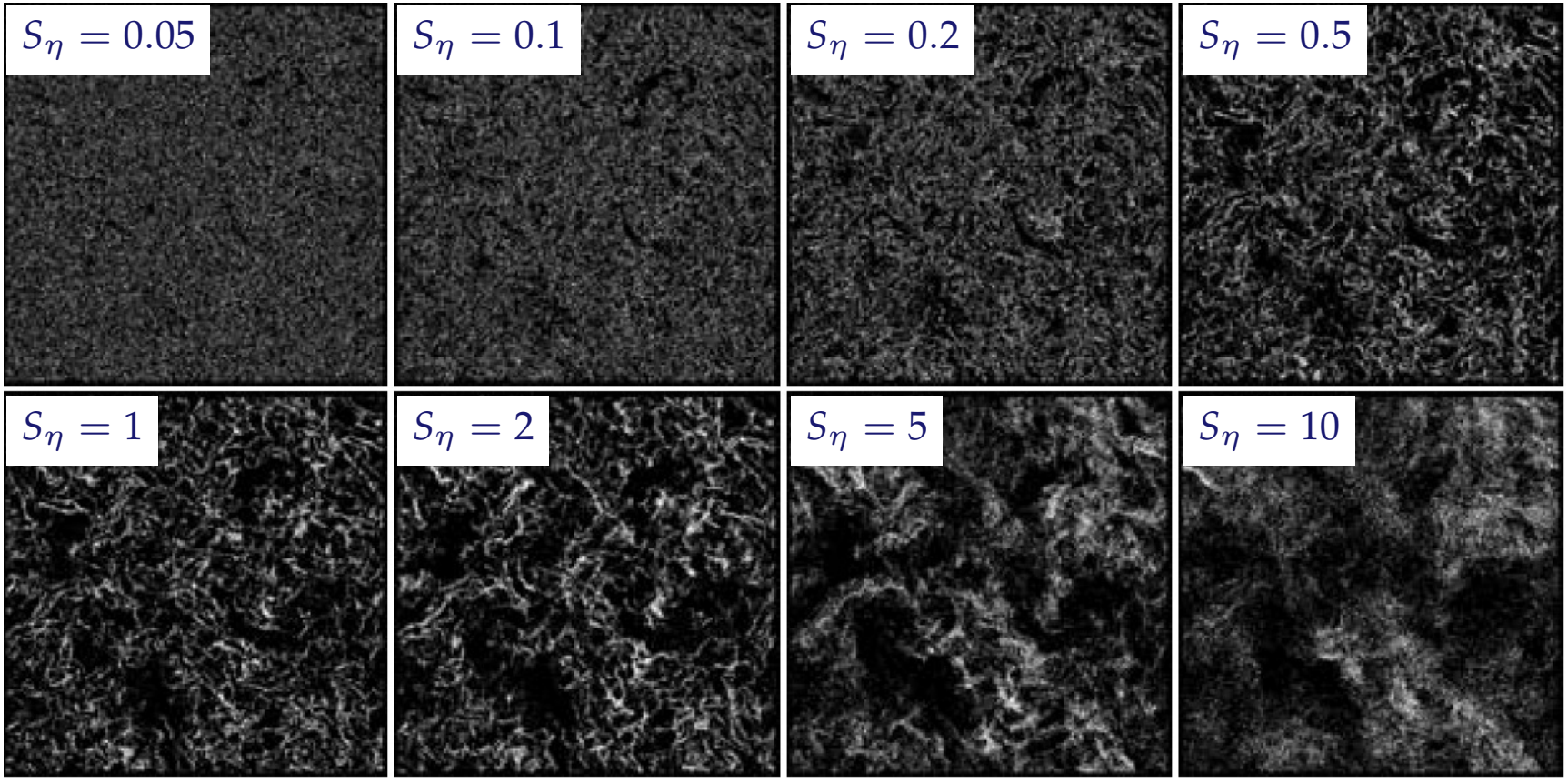
Collision alteration



Flow modulation



Preferential concentration and clustering



Yoshimoto et *al*, *JFM*, 577, **2007**

Preferential concentration $\rightarrow$ one particle

Clustering $\rightarrow$ many particles

Clusters $\rightarrow$ inter-particles

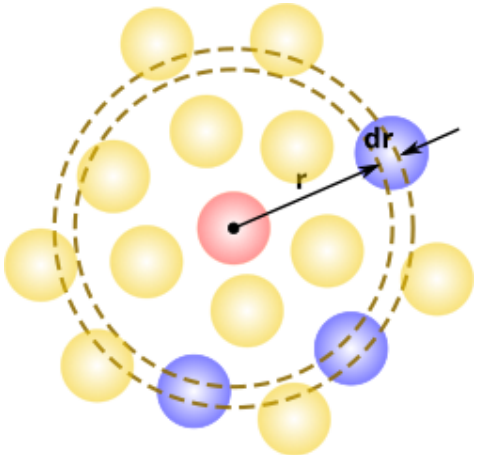
Quantifying clustering

- Box counting
- Radial distribution function
- Voronoï diagrams
- Correlation dimension
- Minkowski functionnals
- …

Quantifying clustering

- Box counting
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- ...

Radial distribution function



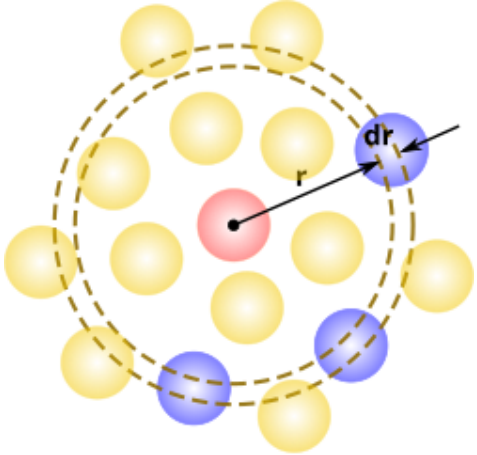
from Wikipedia

- Scale by scale estimation
- Orientation information

Quantifying clustering

- Box counting
- Radial distribution function
- Voronoï diagrams
- Correlation dimension
- Minkowski functionnals
- …

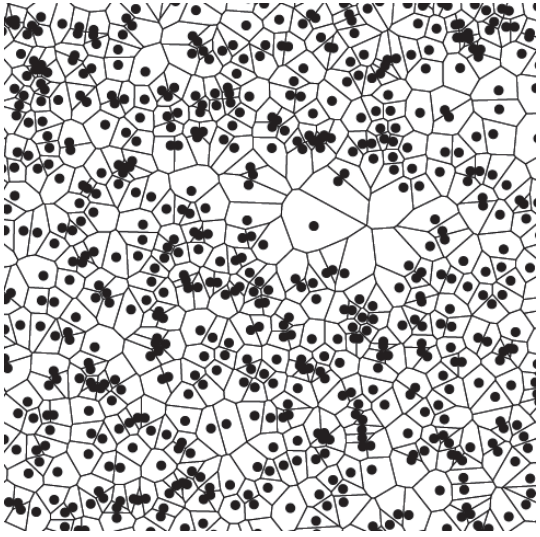
Radial distribution function



from Wikipedia

- Scale by scale estimation
- Orientation information

Voronoï diagrams



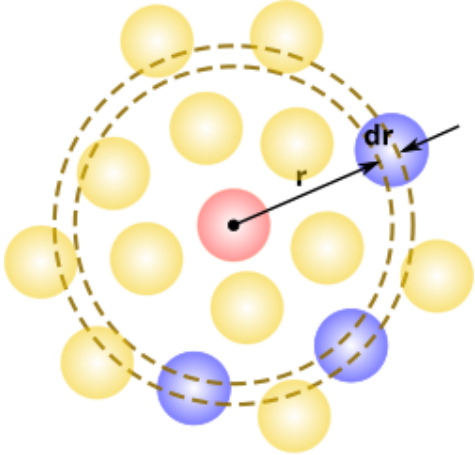
Monchaux, NJP, 2012

- Intrinsic resolution
- Local concentration
- Cluster identification

Quantifying clustering

- Box counting
- Radial distribution function
- Voronoï diagrams
- Correlation dimension
- Minkowski functionnals
- …

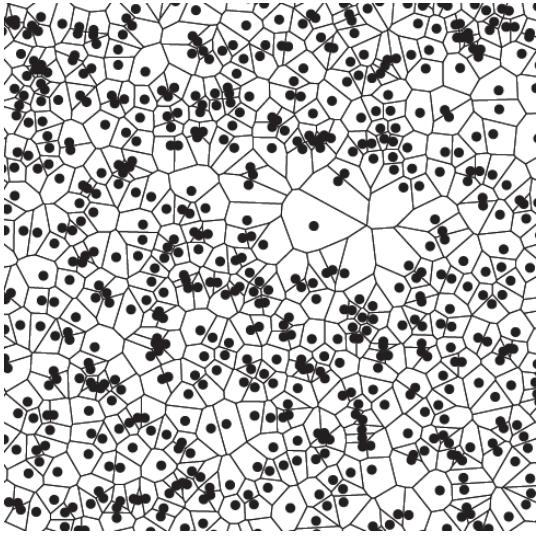
Radial distribution function



from Wikipedia

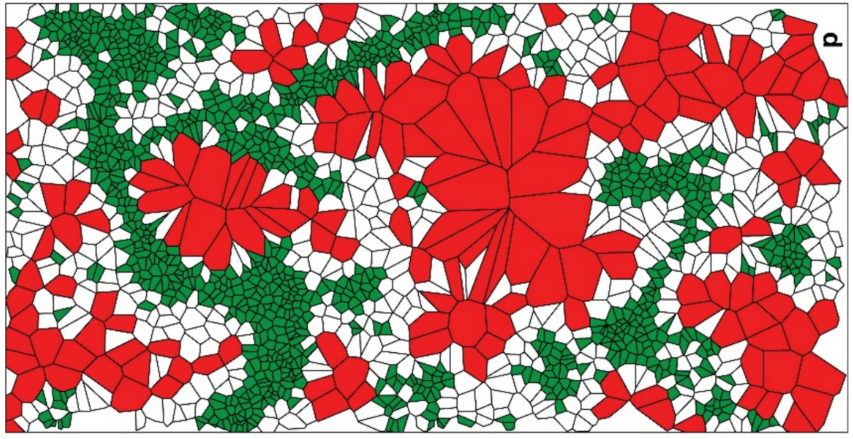
- Scale by scale estimation
- Orientation information

Voronoï diagrams



Monchaux, NJP, 2012

- Intrinsic resolution
- Local concentration
- Cluster identification

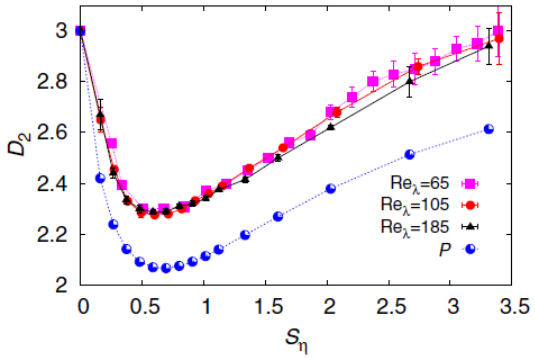


Capponi et al, 2026, Bulletin of volcanology

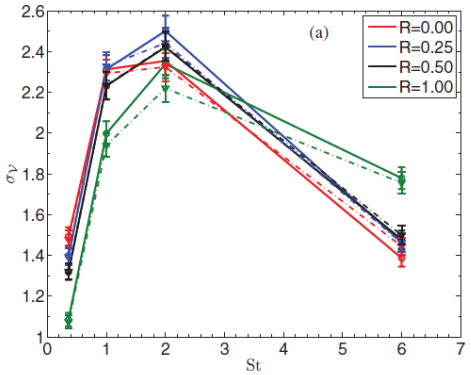
Main results

- Maximal when $St \simeq 1$

Maximal clustering



Bec et al PRL, 2007

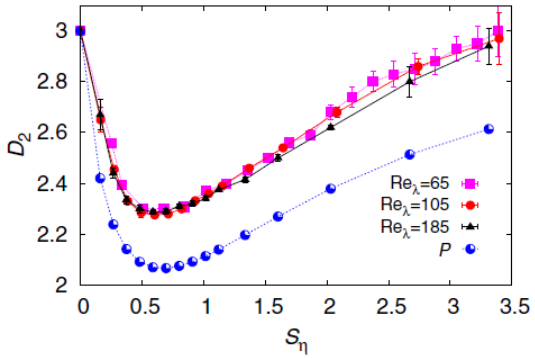


Dejoan & Monchaux, POF, 2013

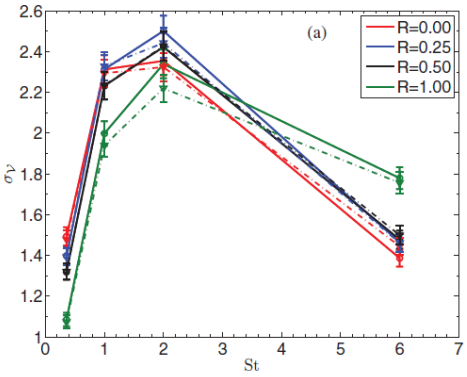
Main results

- Maximal when $St \simeq 1$
- Typical scale around 10η

Maximal clustering

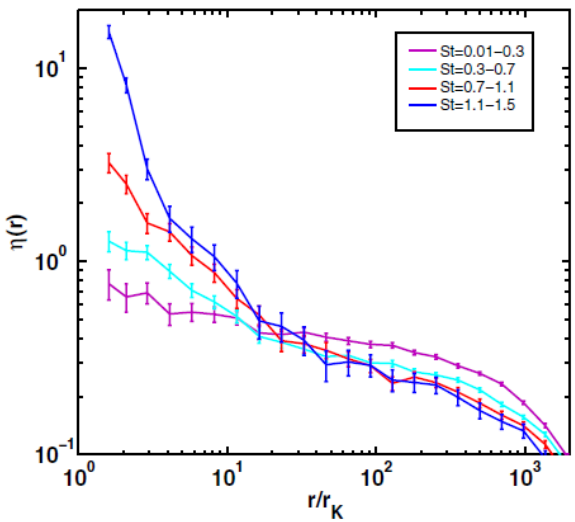


Bec et al PRL, 2007



Dejoan & Monchaux, POF, 2013

Typical size around 10η

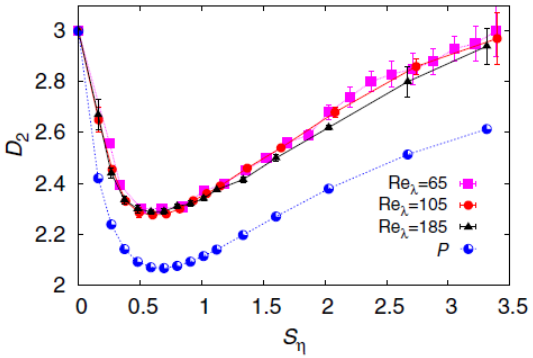


Saw et al, PRL, 2008

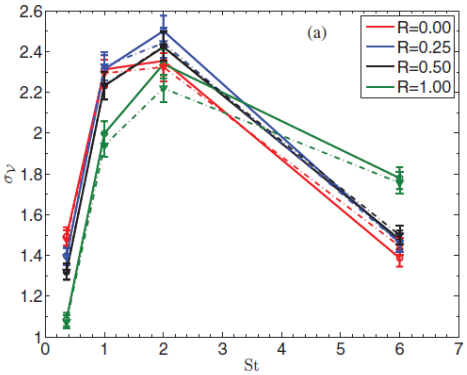
Main results

- Maximal when $St \simeq 1$
- Typical scale around 10η
- Shear and gravity : anisotropy

Maximal clustering

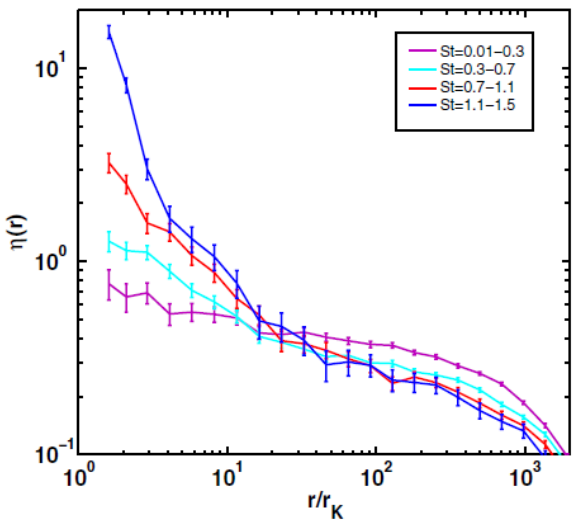


Bec et al PRL, 2007



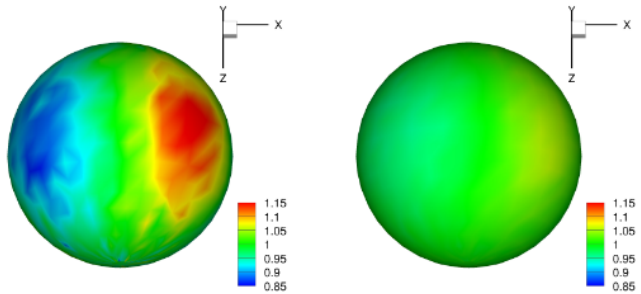
Dejoan & Monchaux, POF, 2013

Typical size around 10η



Saw et al, PRL, 2008

Anisotropy at small scales

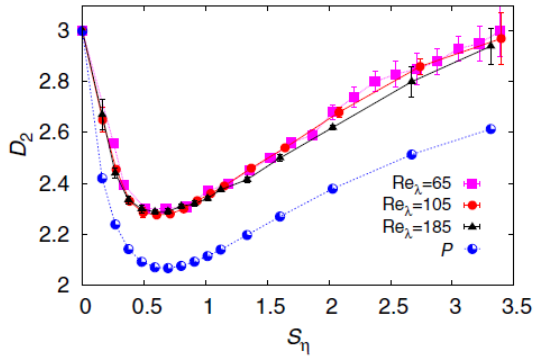


Gualtieri et al, JFM, 2009

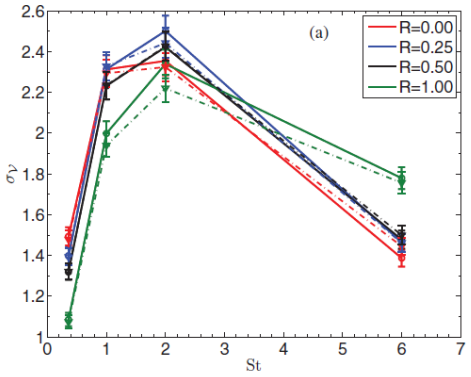
Main results

- Maximal when $St \simeq 1$
- Typical scale around 10η
- Shear and gravity : anisotropy
- Scale invariant at low St
- Multi scale phenomenon

Maximal clustering

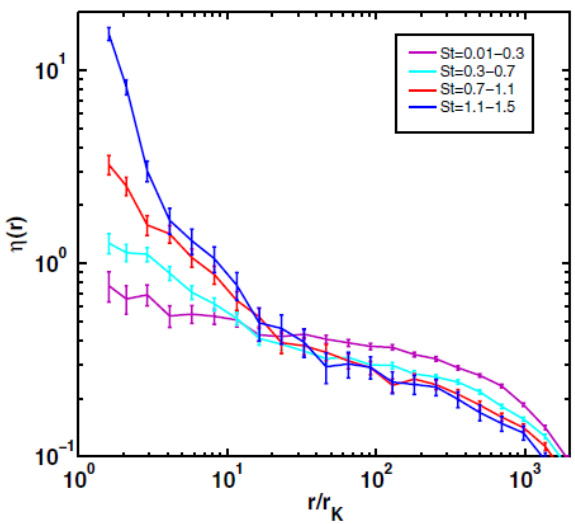


Bec et al PRL, 2007



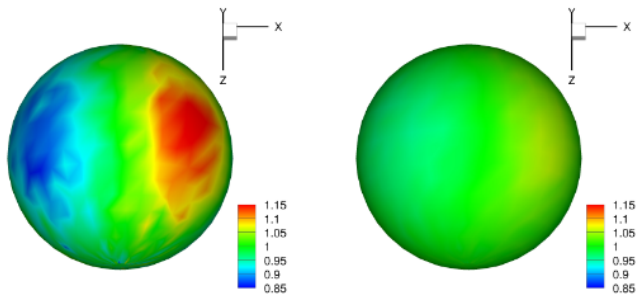
Dejoan & Monchaux, POF, 2013

Typical size around 10η



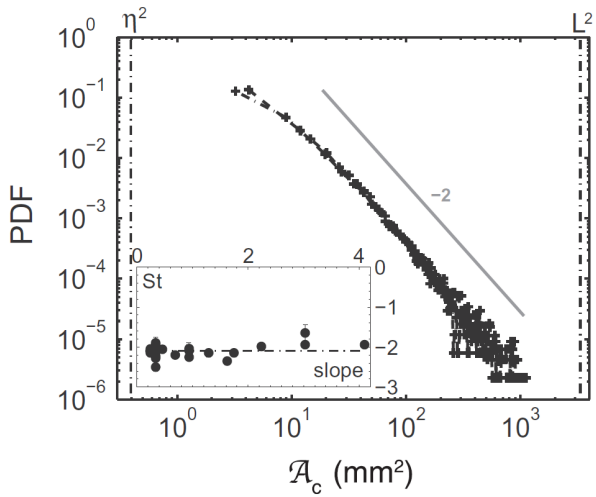
Saw et al, PRL, 2008

Anisotropy at small scales



Gualtieri et al, JFM, 2009

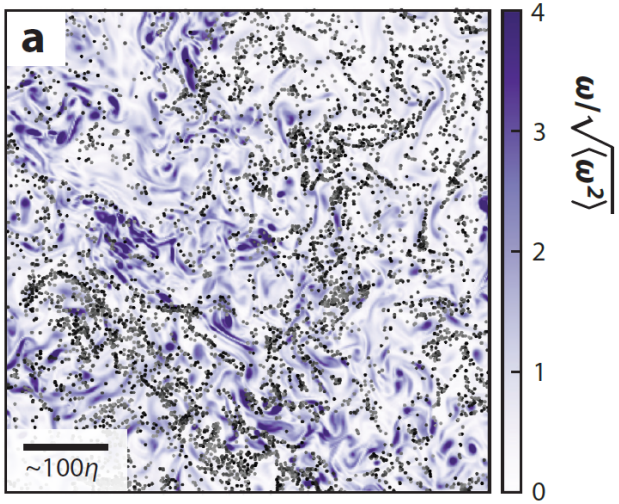
Self similar clustering



Monchaux et al, POF, 2008

Mechanisms

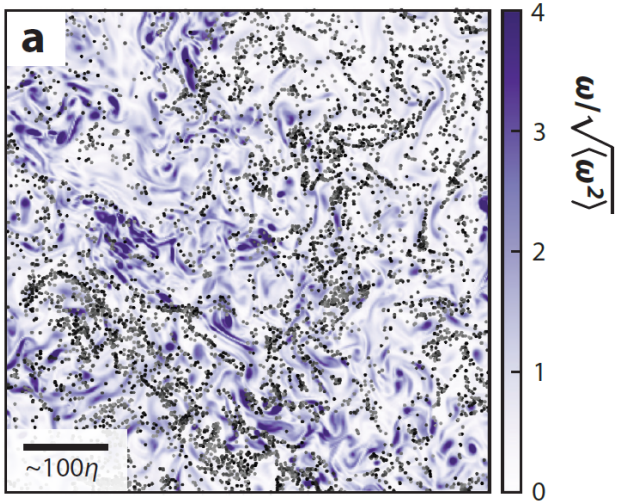
Centrifugal effect



Bec et al ARFM, 2024

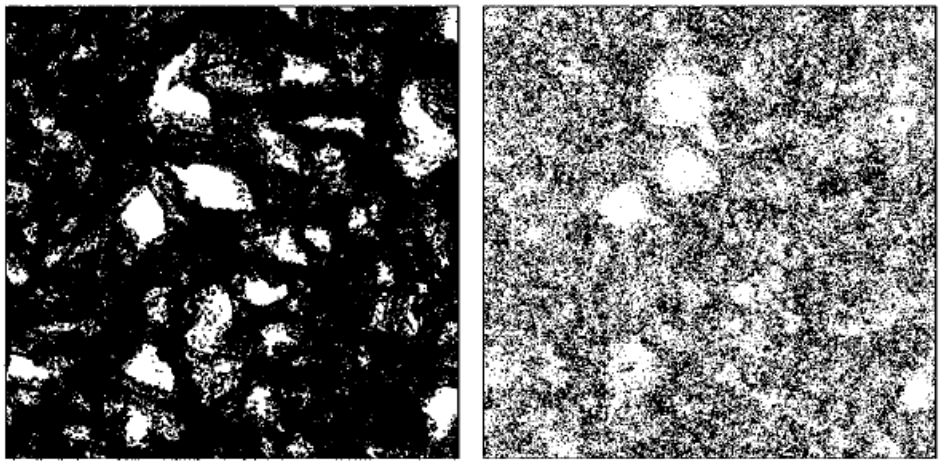
Mechanisms

Centrifugal effect



Bec et al ARFM, 2024

Sweep-stick mechanism

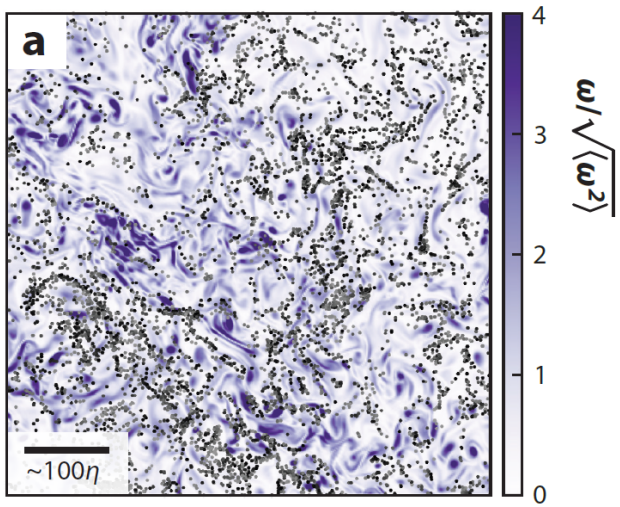


Coleman & Vassillicos, POF, 2009

- Particles stick to zero acceleration points
- Zero acceleration points are swept and cluster

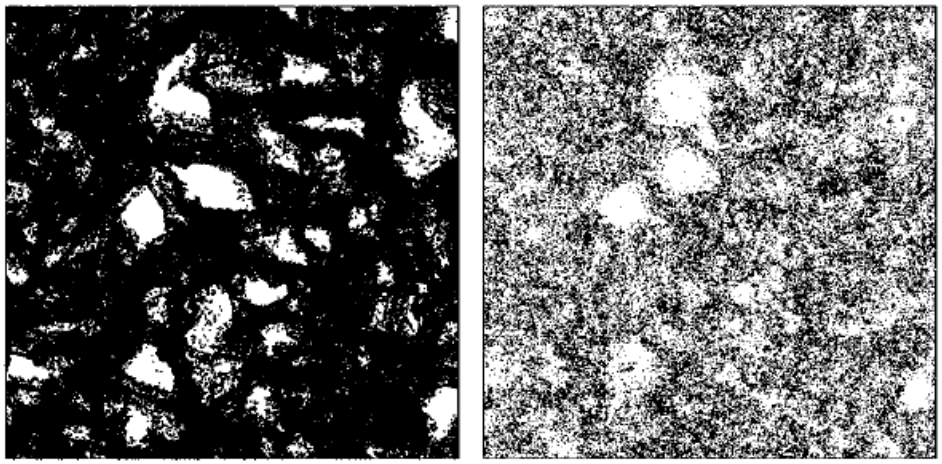
Mechanisms

Centrifugal effect



Bec et al ARFM, 2024

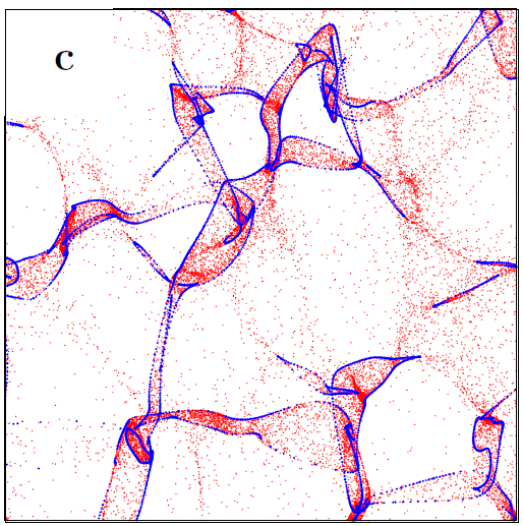
Sweep-stick mechanism



Coleman & Vassillicos, POF, 2009

- Particles stick to zero acceleration points
- Zero acceleration points are swept and cluster

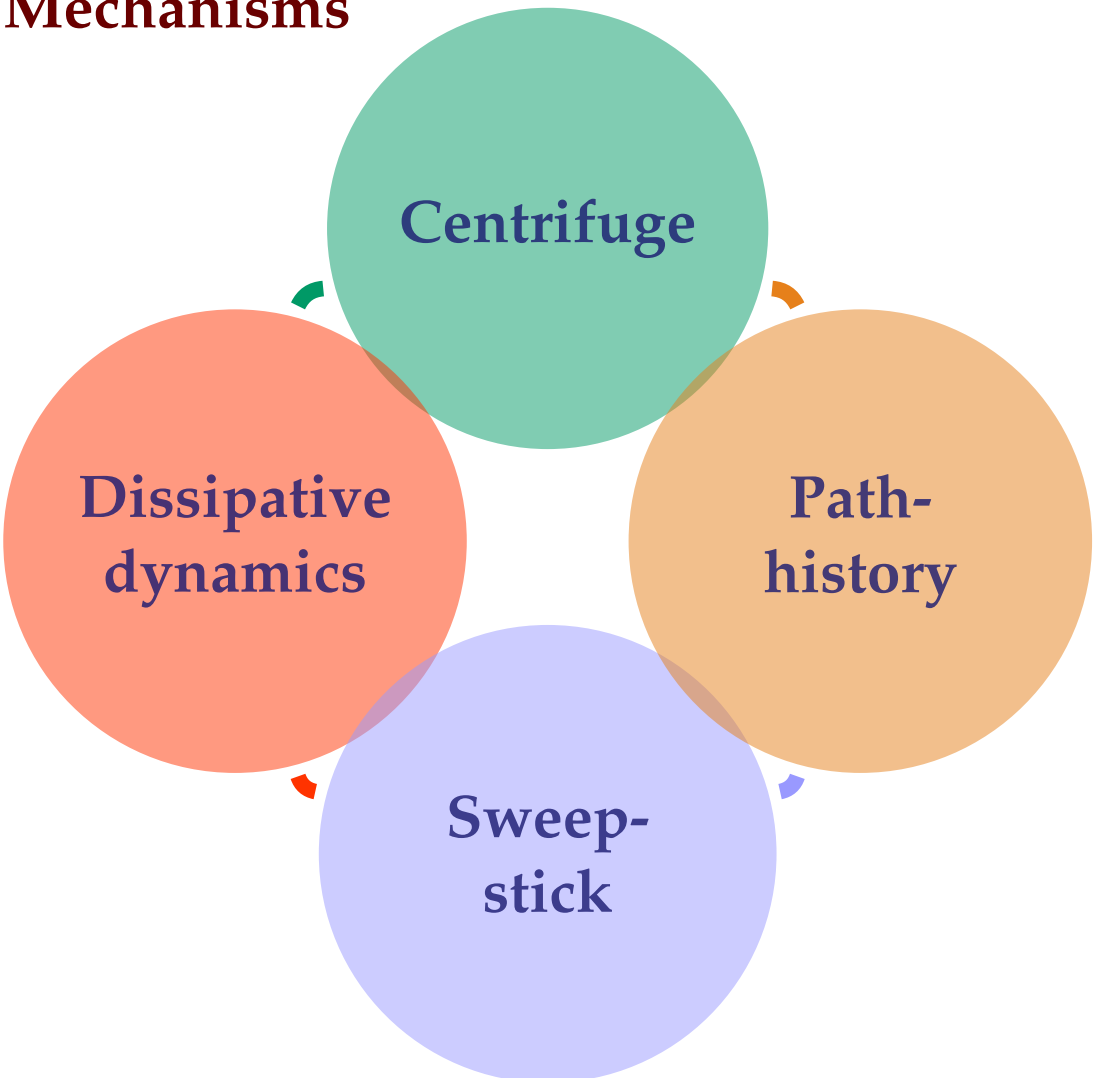
Dissipative dynamics



Wilkinson & Mehlig, EPL, 2005

- Compressible particle field
- Phase space contraction
- Accumulation on caustics

Mechanisms



Local mechanisms :

Heavy particles ejected from vortices
Particles live in high strain regions
Low Stokes number limit
Maxey and followers

Particles stick to 0-acceleration points
0-acceleration points form clusters
High R_λ and low Stokes limits
Debated
Vassillicos and collaborators

Non local mechanisms (time) :

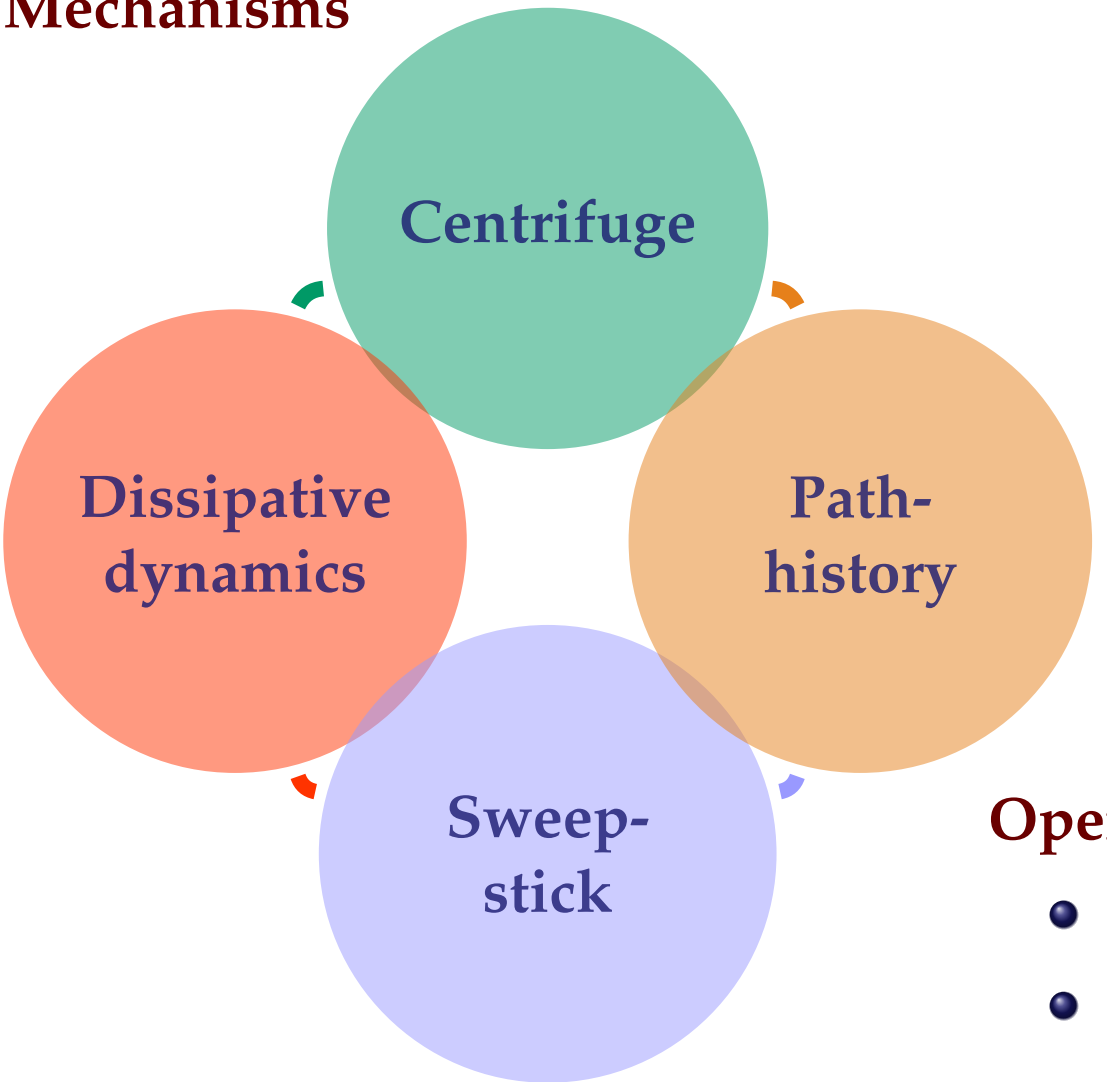
Path history of particles matters
Symmetry breaking
Unify inertial and dissipative regimes
Encompass sweep-stick
Bragg and Collins

Non local mechanisms (space) :

Clusters without preferential concentration
Phase space contraction
Caustics formation

Bec, Gustvasson and Melhig, see Annual Review, 2024

Mechanisms



Open questions :

- Mechanism hierarchy
- Reynolds number dependency
- Gravity effects
- Two-way coupling
- Finite size